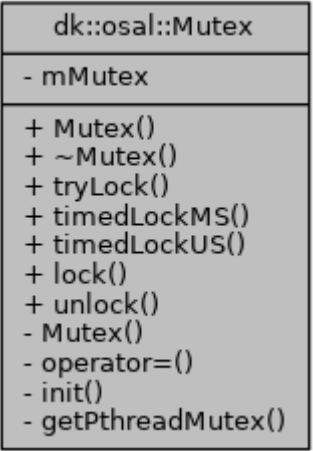


dk::osal::Mutex Class Reference

A mutual-exclusion lock. Only one thread may have a **Mutex** locked at any one time. [More...](#)

#include <DKOsal**Mutex.h**>

Collaboration diagram for dk::osal::Mutex:



Public Member Functions

Mutex ()	Construct a new Mutex . More...
~Mutex ()	Destructor. Note that destroying a locked mutex can lead to deadlock. More...
bool tryLock ()	Try to lock the Mutex , The Mutex is locked if it is not locked by another user. More...
bool timedLockMS (uint32_t msec)	The Mutex . (Waits for a timeout) Blocks the current thread until the lock is acquired, or the specified timeout milliseconds duration interval elapsed. More...
bool timedLockUS (uint32_t usec)	Lock the Mutex . (Waits for a timeout) Blocks the current thread until the lock is acquired, or the specified timeout microseconds duration interval elapsed. More...
void lock ()	lock the mutex. OsalMutex component shall provide APIs to lock the critical section using mutex lock functionality More...
void unlock ()	unlocks the mutex. OsalMutex component shall provide APIs to unlock the critical section More...

Private Member Functions

Mutex (const Mutex &)	Disable Copy constructor and Assignment operator on Mutex . More...
Mutex & operator= (const Mutex &)	
void init ()	Initialize the Mutex . More...
pthread_mutex_t * getPthreadMutex ()	Get pointer to internal pthread_mutex. More...

Private Attributes

pthread_mutex_t **mMutex**

Detailed Description

A mutual-exclusion lock. Only one thread may have a **Mutex** locked at any one time.

requirement ID 751534

Constructor & Destructor Documentation

◆ **Mutex()** [1 / 2]

dk::osal::Mutex::Mutex ()

Construct a new **Mutex**.

requirement ID 751559

◆ **~Mutex()**

dk::osal::Mutex::~~Mutex ()

Destructor. Note that destroying a locked mutex can lead to deadlock.

requirement ID 751559

◆ **Mutex()** [2 / 2]

dk::osal::Mutex::Mutex (const **Mutex** &)

Disable Copy constructor and Assignment operator on **Mutex**.

Member Function Documentation

◆ **getPthreadMutex()**

pthread_mutex_t * dk::osal::Mutex::getPthreadMutex ()

private

Get pointer to internal pthread_mutex.

requirement ID 751534

◆ init()

dk::osal::Mutex::init ()

private

Initialize the **Mutex**.

requirement ID 751559

◆ lock()

void dk::osal::Mutex::lock ()

lock the mutex. OsalMutex component shall provide APIs to lock the critical section using mutex lock functionality

requirement ID 751563

Here is the call graph for this function:

Here is the caller graph for this function:

◆ operator=()

Mutex& dk::osal::Mutex::operator= (const **Mutex** &)

private

◆ timedLockMS()

dk::osal::Mutex::timedLockMS (uint32_t msec)

The **Mutex**. (Waits for a timeout) Blocks the current thread until the lock is acquired, or the specified timeout milliseconds duration interval elapsed.

Parameters

msec The number of milliseconds to wait before the **lock()** is cancelled

Returns

true lock acquired false timeout happened

requirement ID 751561

Here is the caller graph for this function:

◆ timedLockUS()

dk::osal::Mutex::timedLockUS (uint32_t usec)

Lock the **Mutex**. (Waits for a timeout) Blocks the current thread until the lock is acquired, or the specified timeout microseconds duration interval elapsed.

Parameters

usec The number of microseconds to wait before the **lock()** is cancelled

Returns

true lock acquired false timeout happened

requirement ID 751562

Here is the caller graph for this function:

◆ tryLock()

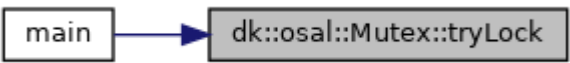
dk::osal::Mutex::tryLock ()

Try to lock the **Mutex**, The **Mutex** is locked if it is not locked by another user.

Returns
true lock acquired **false** lock is held by another user.

requirement ID 751560

Here is the caller graph for this function:



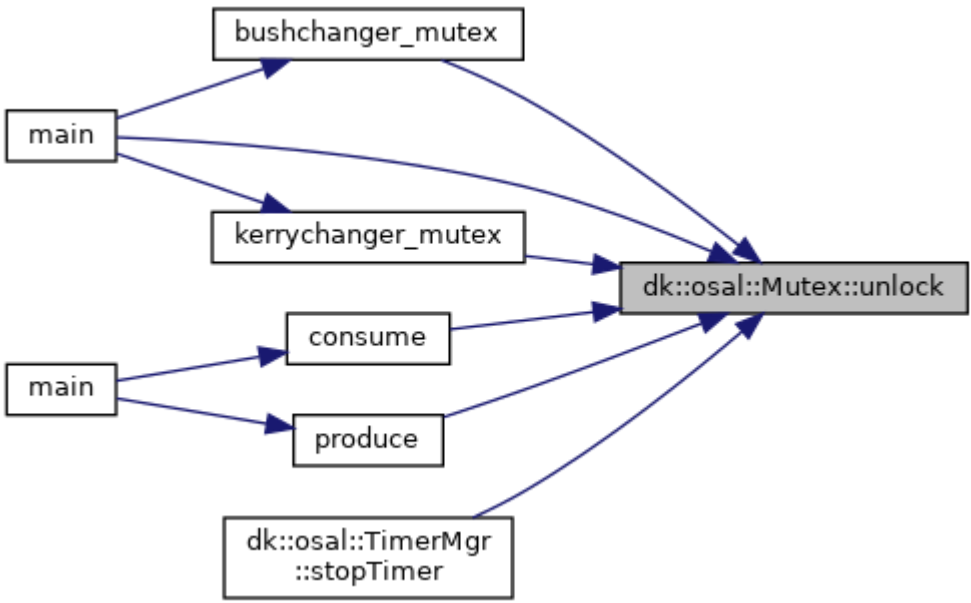
◆ unlock()

void dk::osal::Mutex::unlock ()

unlocks the mutex. OsalMutex component shall provide APIs to unlock the critical section

requirement ID 751564

Here is the caller graph for this function:



Member Data Documentation

◆ mMutex

pthread_mutex_t dk::osal::Mutex::mMutex

private

The documentation for this class was generated from the following files:

- /home/damudhar/29_8_audi/cluster-platform/dijkstra/abstraction/osal/src/posix/**DKOsalMutex.h**
- /home/damudhar/29_8_audi/cluster-platform/dijkstra/abstraction/osal/src/posix/**DKOsalMutex.cpp**